

DN

# DARIUS MIRCEA NISTOR

nistordarius26h@gmail.com | +40787726697 | WWW: <https://nddev.dpdns.org>

## Summary

---

Pre-university student from Braşov, Romania, with hands-on experience across mechatronics, embedded systems, PCB design, and CAD engineering, backed by self-built projects including a global-accessible autonomous robot, custom RF research hardware, and AI-assisted computer vision systems - each taken from schematic/code through to a working physical build, debugged end to end. Uses AI tools as a development accelerant. Seeking an internship or part-time role compatible with school hours in mechatronics, electronics, or related engineering fields.

## Skills

---

- Mechatronics & Embedded Systems (ESP32, Raspberry Pi, Arduino)
- PCB Design (EasyEDA)
- CAD / 3D Design (SolidWorks, Fusion 360, FreeCAD)
- Circuit Design & Electronics
- Python
- AI-Assisted Development
- Git & GitHub
- Linux / Bash
- Data Analysis & Visualization
- Cybersecurity (Wireless/RF protocol analysis)

Additional skills listed on [LinkedIn](#)

---

## Education and Training

---

Colegiul Național de Informatică "Grigore Moisil", Braşov  
Mathematics & Informatics track, pre-university student

Relevant Coursework: Advanced Mathematics,  
Computational Thinking, Physics, Electronics Fundamentals.

## Languages

---

Romanian: Native

English: Upper Intermediate (B2 - C1 EF SET Certified)

German: Intermediate (B1)

## Certifications & Licenses

---

- Diploma in Mechatronics (Alison)
- CS50's Introduction to Programming with Python (Harvard University)
- Robotics and AI Certification (Great Learning)
- Google AI Professional Certificate
- Electrical Engineering Certification (Cursa)
- Google AI Essentials Specialization Certificate
- AWS Concepts Certification (Amazon Web Services)
- Risk Management Course (The Open University & Rolls-Royce)
- Basics of Cyber Law (Cambridge International Qualifications)
- EF SET English Certification – C1 Advanced

Additional certifications listed on [LinkedIn](#)

---

## Websites & Social Links

---

- <https://nddev.dpdns.org> — Personal Portfolio
- [www.linkedin.com/in/darius-nistor-3292783b1](https://www.linkedin.com/in/darius-nistor-3292783b1) — LinkedIn
- <https://github.com/nistordarius26h-ship-it> — GitHub

## Projects

---

- Globally Accessible Autonomous Robotics Platform

Built a robot that can be controlled from anywhere in the world over a cellular connection — an ESP32 handles real-time motor control and sensor polling, while a Raspberry Pi runs the higher-level orchestration, video streaming, and AI workload. Managed an important hardware & software flaw; traced it to a loose contact pin on the PCA controlling the four motor drivers - each time it made intermittent contact, it caused signal corruption from the microcontroller to the PCA resulting in false signals to the motors. Also went through two LTE modems — the first one failed to authenticate with the provider's cellular tower, so switched to a more capable one. Getting remote video to work reliably meant solving carrier-grade NAT traversal, which led to standing up a self-hosted TURN server for WebRTC. Designed the chassis in SolidWorks and Fusion 360, prioritizing easy maintenance, then built and tested a physical prototype. The sensor suite includes ultrasonic proximity detection, acoustic command triggers, and temperature/humidity/rain sensing, all feeding into a telemetry system that reports back to the operator live.

→ [https://github.com/nistordarius26h-ship-it/ga\\_ai\\_robot](https://github.com/nistordarius26h-ship-it/ga_ai_robot)

- Real-Time Computer Vision & Autonomous Navigation Suite

Built a real-time computer vision pipeline in Python combining a YOLO11x detection model with ByteTrack, so it can follow multiple objects at once even when they're partially blocked from view, while computing motion trails to predict where each one is headed. Frames started piling up because the camera's FPS didn't match the model's inference speed — fixed it; synchronized the live stream frame rate with the model's inference speed. Getting GPU acceleration working was its own battle: built for CUDA first, then had to port it to ROCm for AMD hardware, which took a lot more troubleshooting than expected given how much thinner the tooling is compared to Nvidia's ecosystem. Validated the whole thing against footage from a personal aerial drone, then adapted it again to run on the robot's live camera feed — different source, different latency, different problems to iron out.

→ [https://github.com/nistordarius26h-ship-it/ai\\_multiobject\\_python\\_tracker](https://github.com/nistordarius26h-ship-it/ai_multiobject_python_tracker)

- Custom RF Wireless Security Hardware (PCB Design)

Designed, routed, and fabricated a multi-layer PCB around an ESP32 for wireless protocol research. First revision worked right out of the gate — every connection and circuit performed as designed, no rework needed. The schematic centers on a long-range RF transceiver: careful component selection, decoupling capacitors placed to cut noise, current-limiting resistors to protect the antenna. Spent time optimizing the layout specifically for signal integrity at range, since that's where cheap boards usually fall apart. Handled the whole manufacturing side too — BOM, sourcing, fabrication, and assembling the SMD components by hand. On the firmware side, wrote embedded code to probe Wi-Fi, Bluetooth, and BLE behavior, running passive RF monitoring and protocol analysis to find security gaps in how these devices actually communicate.

→ <https://github.com/nistordarius26h-ship-it/esp32jamm>

- AI-Based Aircraft Stall Margin Prediction System (Research Prototype)

Set out to see whether machine learning could flag an approaching aerodynamic stall earlier than standard stall-warning systems do. Spent a lot of time on the research side — analyzing airflow disturbance patterns against flight telemetry to build a predictive model — even though the project itself, being research-only, didn't hit the kind of hardware failures the physical builds did. Also explored fusing forward-looking LiDAR with onboard telemetry, aiming for a system that could assess stall risk proactively instead of reacting after the fact.

→ <https://github.com/nistordarius26h-ship-it/AI-Stall-Prediction-System>

- Automated Quantitative Trading System

Built an algorithmic trading system for the XAU/USD market that runs without any manual input — market data analysis, signal generation, and trade execution all handled automatically. Getting it to actually turn a profit outside of demo conditions took a lot of trial and error; strategies that looked solid on paper or in a demo account didn't always hold up once real spreads, slippage, and live volatility got involved. Built in automated risk management so the account stays protected even when the market moves against the strategy.

→ Currently deployed and operating on my personal live trading account

- Portable Local AI Workspace

Set up a fully offline, USB-portable AI development environment for privacy-focused work — no cloud dependency, everything runs locally. Originally built it for Linux, then adapted the setup to run on Windows too, which took some extra work to get the local inference stack behaving the same way across both. Tuned local inference for Gemma, Qwen & many more models so the assistant stays fully functional with nothing leaving the device.

→ <https://lnkd.in/p/eeeWfmbi>

## Awards & Honors

---

- InfoTron 2022 – 1st Place (Mechanical Engineering, Electronics & Computer Science Competition sponsored by Miele Braşov and Siemens Industry Software)
- InfoTron 2023 – 3rd Place (Mechanical Engineering, Electronics & Computer Science Competition)
- InfoTron 2026 – Participant

## Activities

---

- Competitive Swimming Athlete (2012–2020) - Won 50+ regional and national medals.